

TIME: 3 HRS

MPIHARM
(SEM. II) THEORY EXAMINATION 2025-26
MEDICINAL PLANT BIOTECHNOLOGY

M.MARKS: 75

Note: Attempt all Sections.

SECTION A

1. Attempt all questions in brief.

10 x 2 = 20

a.	Define plant biotechnology.
b.	What is recombinant DNA technology?
c.	Define gene expression.
d.	What is micropropagation?
e.	Define somaclonal variation.
f.	What is protoplast fusion?
g.	Define immobilization of plant cells.
h.	What is elicitor?
i.	Define transgenic plant.
j.	What is PCR?

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

a.	Explain genetic engineering techniques in medicinal plants and their applications.
b.	Discuss tissue culture techniques including organogenesis, embryogenesis and micro propagation.
c.	Describe immobilization techniques and their role in secondary metabolite production.

SECTION C

3. Attempt any five of the following:

7 x 5 = 35

a.	Write a note on historical development and applications of plant biotechnology.
b.	Discuss DNA, RNA and protein synthesis in relation to gene expression.
c.	Explain sterilization methods used in plant tissue culture.
d.	Write short note on hairy root culture and its applications.
e.	Discuss role of precursors and elicitors in production of secondary metabolites.
f.	Explain bioreactors used for large-scale plant cell culture.
g.	Write a note on PCR and its applications in plant genome analysis.